Kushagra Tiwary

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Research Statement

I am a PhD Student in Ramesh Raskar's group focused on AI-based computational discovery specifically in the context of vision. I also work with Brian Cheung and Tomaso Poggio. I am using my research to focus on two key questions: 1) How can embodied agents trained via deep reinforcement learning (DRL) be used as "what-if" or hypothesis testing machines? 2) How can we use learned simulators with DRL to find interesting artifacts and create better AI systems? Some Thoughts.

Education

- **Massachusetts Institute of Technology** — Cambridge, MA
  *PhD in Media Arts & Sciences, **Camera Culture**, Media Lab* — *Sep 2023 - Present*
  Advisor: (**Ramesh Raskar**)
- **Massachusetts Institute of Technology** — Cambridge, MA
  *SM in Media Arts & Sciences, **Camera Culture**, Media Lab* — *Jul 2021 - May 2023*
  Advisor: (**Ramesh Raskar**)
- **University of Illinois, at Urbana-Champaign** — Champaign, IL
  Bachelors with Honors in Electrical and Computer Engineering — *Aug 2015 - May 2019*

Selected Honors & Awards

* **Impact papers on Generative AI $70,000**: Invited to submit a perspective piece on the impact of Generative AI on science and engineering to MIT Press ($\frac{1}{16}$ proposals selected across MIT) **MIT News**
* **2023 North America Winner of the Qualcomm Innovation Fellowship**: **Fellowship Recipient List**, **Media Lab News**, **EECS News**
* **MISTI MIT-Israel Zuckerman STEM Fund Award $30,000**: **MIT-Israel Zuckerman STEM Fund Award** (one of six proposals selected across MIT)

Selected Invited Talks & Workshops

* **Oral at NECV 2024- New England Computer Vision Conference**: What if Machines for Vision: Computationally Emulating the Evolution of Visual Intelligence
* **Talk at MIT BCS- Tomaso Poggio's Group**: Computational Evolution for Testing Vision Hypotheses, October 2024
* **UCR Graduate Semiar Series**: Computational Evolution of Visual Intelligence, September 2024
* **Organizer for ECCV Workshop on Neural Fields Beyond Conventional Cameras**: Workshop accepted at the European Conference on Computer Vision (ECCV) in Milan, Italy (ECCV Attendance: 10k+), September 2024
* **Organizer: AI For Accelerating Scientific Discovery**: Using AI to Accelerate Science, R&D, and Augment Engineering & Design. **slides**
* **CSAIL Graphics Seminar**: Neural Rendering and Secondary Cues: Learning Hidden Neural Radiance Fields using Reflections and Shadows, **slides**, video upcoming!

Selected Research Experience

- **Camera Culture, MIT Media Lab** — Cambridge, MA
  Graduate Research Assistant — *Jul 2021 - Present*
  - **Ph.D. Student**: My work broadly focuses on AI Based Computational Discovery *i.e.* how can we study and discovery new things about the world?
  - **Masters Student**: Thesis Title: *Discovering, Learning & Exploiting Visual Cues*: How can modern data-driven frameworks exploit physics-based cues to observe the hidden and invisible parts of the scene?
    Thesis Committee: **Ramesh Raskar** (**Advisor**, Prof. MIT), **Pulkit Agarwal** (Prof. MIT), **Fadel Adib** (Prof. MIT)

Selected Publications I've been thinking about lately (See **G. Scholar** for full List)

Computational Discovery

Kushagra Tiwary*, Aaron Young*, ... Tomaso Poggio, Dan-Eric Nilsson, Brian Cheung*, Ramesh Raskar* "What if Eye? Computationally Emulating the Evolution of Visual Intelligence", **website**, **paper** *Nature Journal, Under Submission*

Kushagra Tiwary et. al, **A Roadmap for Generative Design of Visual Intelligence**, *Published at MIT Press: Impacts of Generative AI, 2024*

Tzofi Klinghoffer*, **Kushagra Tiwary***, Nikhil Behari, Bhavya Agarwal, Ramesh Raskar, DiSER: Designing Imaging Systems with Reinforcement Learning, **website** *Published at ICCV, 2023*,

Computer Vision & Machine Learning

Shayne Longpre, N Singh & M Cherep (Speech Lead), **Kushagra Tiwary (Video Lead)**, ..., Alex Pentland, Sara Hooker, Jad Kabbara "Bridging the Data Provenance Gap Across Text, Speech and Video", *ICLR, 2025*

Zaid Tasneem, Akshat Dave, Abhishek Singh, **Kushagra Tiwary**, Praneeth Vepakomma, Ramesh Raskar, Ashok Veeraghavan DecentNeRFs: Decentralized Neural Radiance Fields from Crowdsourced Images, *Published at ECCV 2024*, **website**,

Kushagra Tiwary*, Akshat Dave*, Nikhil Behari, Tzofi Klinghoffer, Ashok Veeraraghavan, Ramesh Raskar, ORCa: Glossy Objects as Radiance Field Cameras, *Published at CVPR, 2023*, **website**, **pdf** **Press:** *MIT News*

Kushagra Tiwary*, Tzofi Klinghoffer*, Siddharth Somasundaram*, Ramesh Raskar, Physics vs. Learned Priors: Rethinking Camera and Algorithm Design for Task-Specific Imaging, *Published at ICCP 2022*, **pdf**

***** denotes **equal contribution**

Selected Relevant Industry Experience

- **AI Group, Optimus Ride *(Autonomous Vehicle Startup from MIT)*** Boston, MA
  AI/Computer Vision Engineer *Jul'19 - Jul'21*
  - **Lead 2nd Gen. Vision Network Design**: Lead architect for *a giant Multi-Tasking Model* deployed on the nationwide fleet; *Lead release testing* of models on next-gen vehicles in Boston. Wrote MultiTasking Codebase from scratch; decreased training and release time by over 25%. (Patented)
  - **Lead Traffic Light Detection & Estimation for AI Team**: Lead for Traffic Light Design & deployed on next-gen vehicles in Boston & Washington DC. (Patented)
  - **Software 2.0 Framework**: Designed company's first Software 2.0 (*Active Learning framework*) that automatically sampled over **40 different rare-events** from disengagement using a system of Teacher Networks from incoming Vision and Lidar Data across nationwide deployments. (Patented)
  - **Lead Design of Labeling Schema for Perception Stack**: Lead company's first Ground Truth Schema for Perception Tasks with Scale.ai, publicized **here**. The Schema was expanded from **10** classes to over **100** classes and attributes.
  - **Lead expansion of Perception Datasets**: Led Data Collection in Boston & Expanded vision dataset size by over **100x**. Created company's *first* Lidar Dataset.
  - **Sensor Suite**: What sensors to select for a self-driving vehicle? (lidars, cameras, thermal, traffic light etc.)

Selected Press

- **This is where the data to build AI comes from**: **Interview at Globe & Mail**
- **AI-generated video has come a long way**: **MIT Technology Review**
- **Paper *Consent in Crisis* was covered by**: **New York Times, 404 Media, Vox, Yahoo! Finance**
- **Tedx Boston: Can AI Recreate 500 Million Years of Vision Evolution?**: **Tedx Boston**
- **Winner of seed grants awarded to MIT scholars studying impacts of generative AI**: **MIT News (2023)**
- **2023 Qualcomm Innovation Fellowship Recipient**: **via EECS News via Media Lab News**
- **Using reflections to see the world from new points of view**: **MIT News (2023), Front Page of MIT on 05/10**
- **Developing safe and reliable systems with high-quality 3D training data**: **Scale AI, (2020)**

Other

- **Patents:**: Efficient detection of structure and status of traffic lights, Crowd-Sourced Neural Radiance Fields (pending)
- **Programming**: Python, C/C++, Tensorflow 1.x/2+, pytorch, OpenCV, TensorRT, ONNX, gRPC, Docker
- **Languages (Full & Professional Working Proficiency)**: English, Spanish (*lived in Spain for 5 years)*, Hindi
- **Reviewer**: TPAMI'CS, CVPR, International Conference on 3D Vision (3DV'2024), MakeMIT 2022 Judge
- **Mentees**: *Nikhil Behari* (PhD Student), *Bhavya Agarwal* (UROP Supervisor, Undergraduate at MIT), *Chaitanya Kapoor* (Undergraduate at BITS Pilani), *Sheshank Shankar* (Undergraduate at Univeristy of Washington)